

N706-EA

EVK User Guide

Issue 1.0
Date 2024-01-30



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Notice

This document is specifically for N706-EA.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706-EA.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2024-01	Initial draft	Ivy

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

Related Documents

Neoway_N706-EA_AT_Command_Mannual

1 EVK Overview

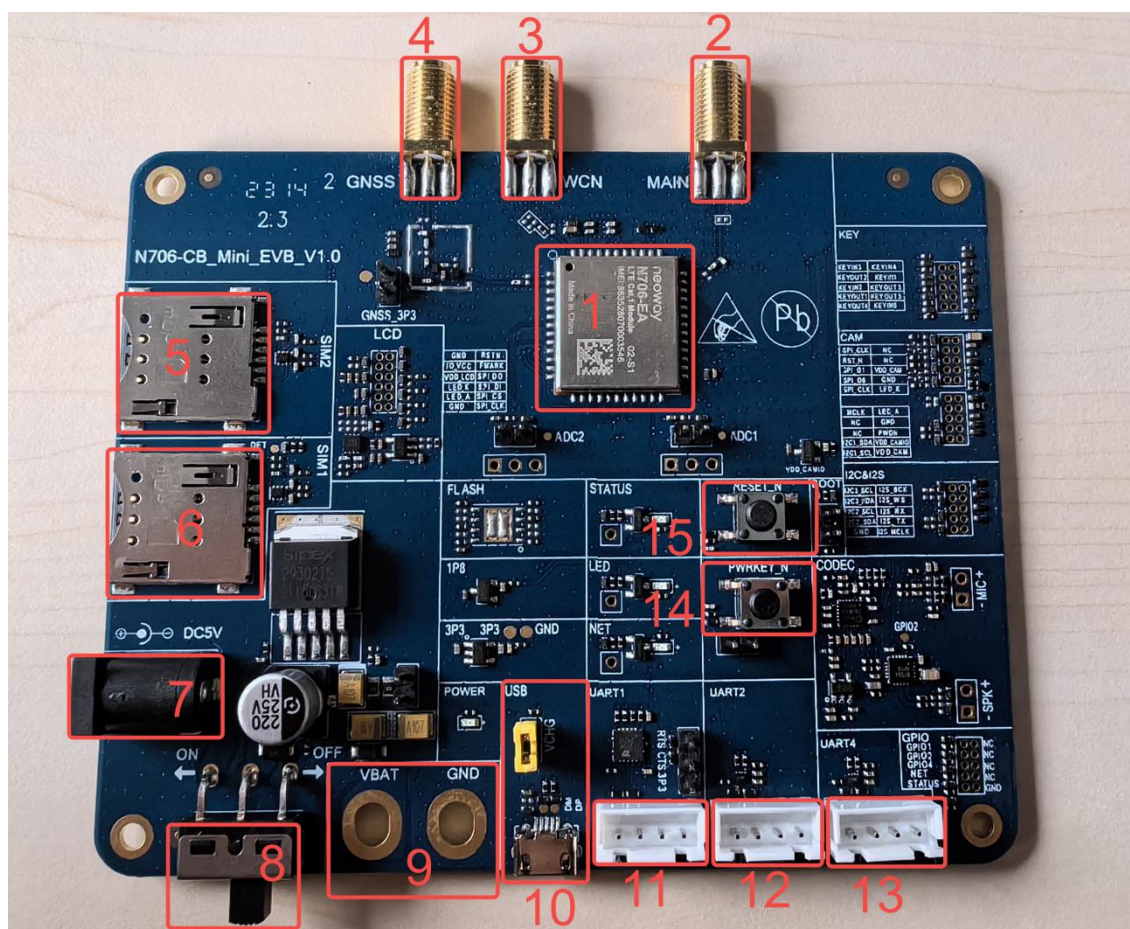
N706 EVB is mainly used for customer software and hardware debugging, offering a variety of peripheral interfaces including power interface, UART interface, SIM card slot, antenna interface, USB 2.0 interface, etc. You can conduct related debugging by connecting the power supply and USB/UART interface.

Available accessories for N706 EVK:

- N706_Mini_EVB_V1.0 (including module)
- M5X0-PWR power serial board
- Others (power adapter, 4G antenna, etc.)

2 EVB Functions

Figure 2-1 Top view of N706-EA EVB



The interface description and functional description of the N706-EA EVB are as shown in the table below:

No.	Button/Interface	Function Description
1	N706 module	Communication module
2	MAIN	Main antenna of the module
3	WCN	WiFi Antenna
4	GNSS	GNSS antenna
5	SIM2	SIM card 2 micro slot (only supported by DG model)
6	SIM1	SIM card 1 micro slot

7	5V power interface	5V DC power input
8	Power switch	"ON" indicates power on, "OFF" indicates power off.
9	Module power	Power positive and negative poles connected to this interface output 3.8 V power supply.
10	USB	Used for upgrading module firmware and capturing module log
11	UART1	Software (Standard) version: Used for AT command communication. Software (Open) version: Used for data transmission.
12	UART2	UART2 can be used to capture the CP side log of the module.
13	UART4	DBG_RXD is only used for Debug debugging. Can be used to capture the AP side log of the module.
14	PWRKEY	Press and release for 1 second to power on (can power on with USB communication)
15	RST	Press and release for 1 second to force the module to restart

3 Power Supply/Connection

The N706 EVB has two power supply methods and communication methods: power supply and power adapter supply. M5X0-PWR serial board or Micro-USB cable is used for data communication. The power supply and communication methods for the EVB are introduced separately below.

3.1 EVB Power Supply

3.1.1 Power Supply with 5V/3A Power Adapter

Figure 3-1 5V/3A power adapter



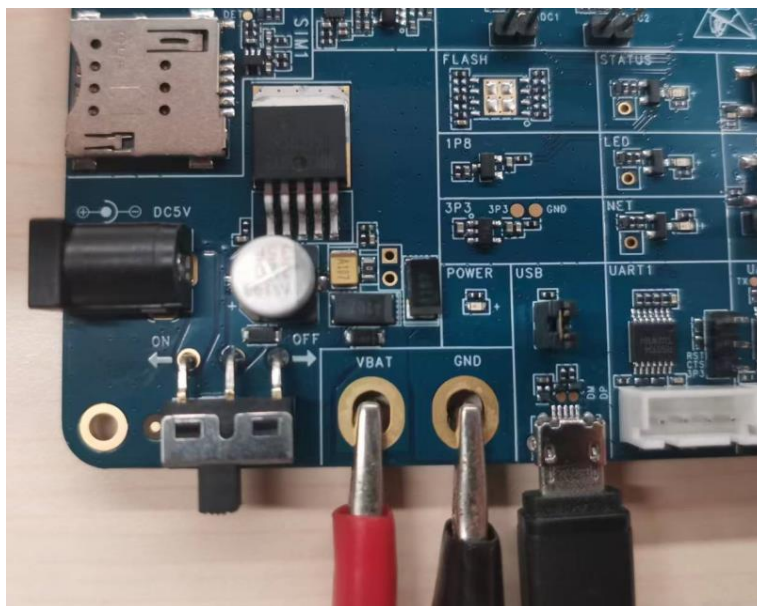
3.1.2 Connecting an External Power Supply

The power supply voltage range for the N706-EA EVB is 3.4V-4.2V. It is recommended to use a regulated power supply outputting 3.8V. Please ensure that the power supply output current reaches at least 2A.



When the N706-EA EVB requires direct power supply to the module through the power pads, please do not reverse the positive and negative poles of the power supply. As shown in the figure above, use the black alligator clip for the negative pole of the power supply, and the red alligator clip for the positive pole.

Reversing the power supply's positive and negative poles or supplying voltage beyond the module's voltage range may damage the module and make it difficult to repair.

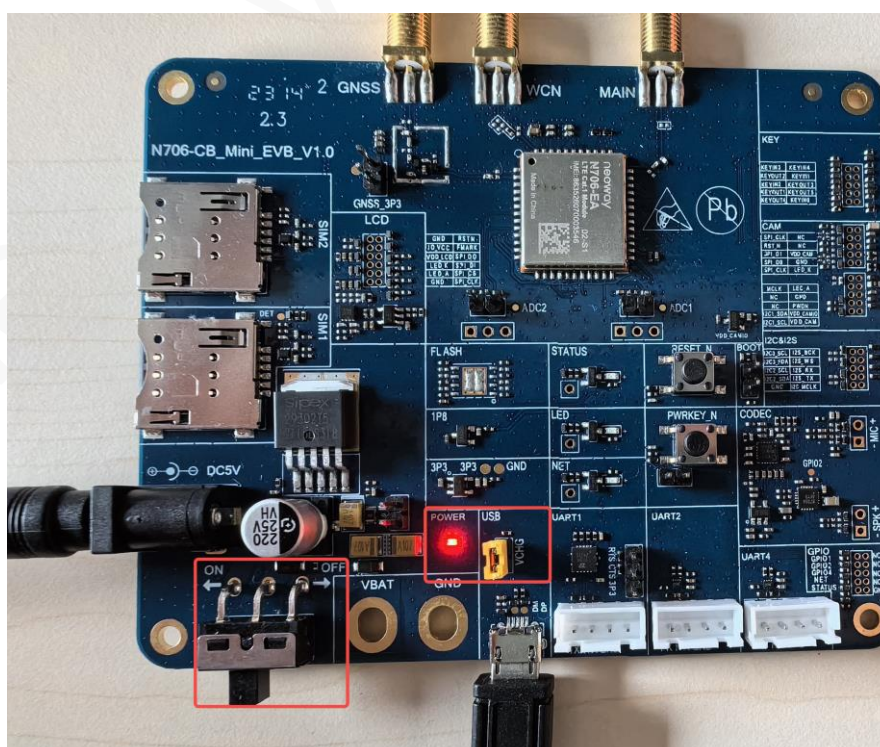


3.2 EVB Power Supply Steps (Taking Power Adapter as Example)

Step 1: Plug the 5V/3A power adapter into the power input interface.

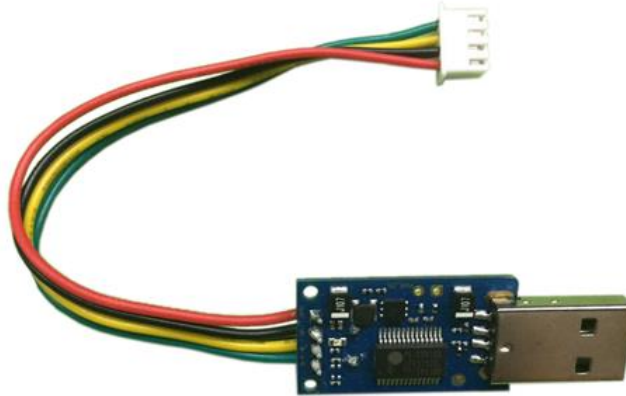
Step 2: Turn on the power switch to the "ON" position.

When the power supply is connected properly to this EVB, the POWER indicator light, as shown in the figure below.



3.3 Communication Connection

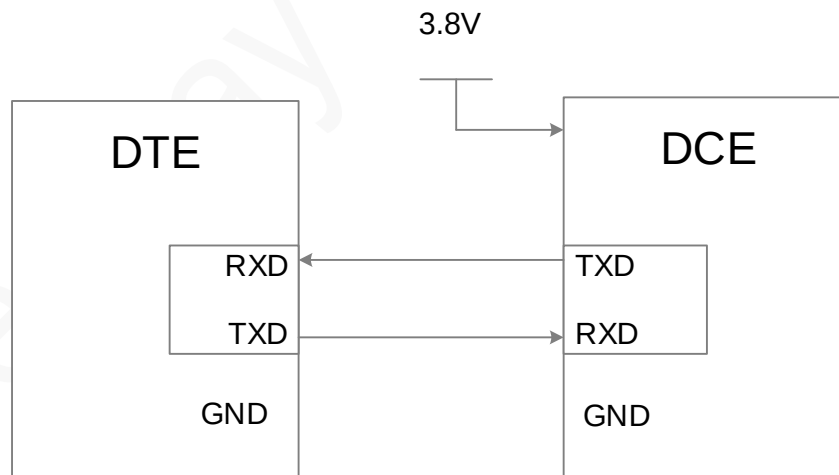
3.3.1 M5X0-PWR Serial Board



The M5X0-PWR serial board outputs 3.3V COMS level through USB to the serial chip, realizing communication between the module and the PC terminal's serial tool through TXD/RXD.

The M5X0-PWR serial port board, designed and produced by our company based on the PL2303 chip, requires the installation of PL2303 drivers before use. Users may also opt for other USB to serial conversion cables/boards based on chips like FT2232, CP210X, etc., and install the respective drivers.

For controlling this module using an MCU, a three-wire connection is required. The connection diagram is as shown in the figure below:



3.3.2 Micro-USB Cable

The micro-USB cable connects the USB interface of the N706-CB_EVB, which can be used for software upgrade, USB communication, and capturing the module's LOG.



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4 Product Debugging

Product debugging can be done through a serial port or USB port. The two methods are introduced separately below:

4.1 Debugging through Serial Port

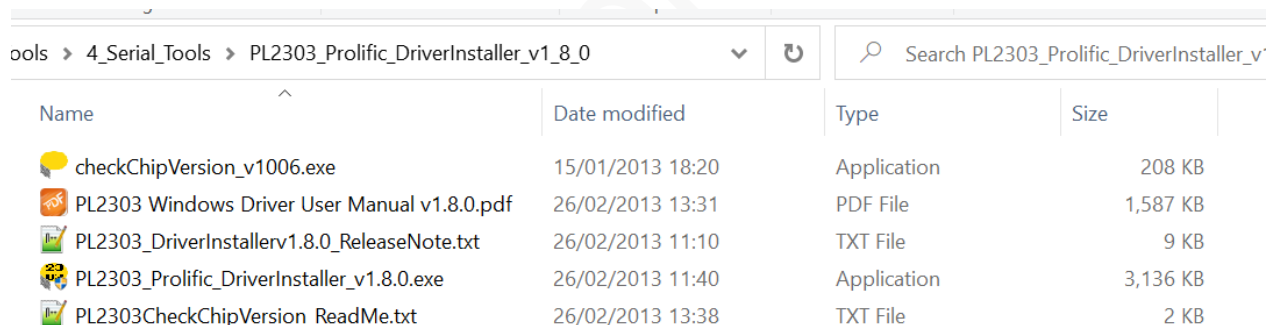
Step 1: Connect the power supply to the N706-EA EVB.

Step 2: Use the M5X0-PWR serial cable to connect the computer and the development board.

Step 3: Power on the module and press the power_key button for 1 second to turn it on normally.

Step 4: Install the serial driver (PL2303).

1. Unzip the serial port driver package. The directory of the unzipped folder is as shown in the figure below.



Name	Date modified	Type	Size
checkChipVersion_v1006.exe	15/01/2013 18:20	Application	208 KB
PL2303 Windows Driver User Manual v1.8.0.pdf	26/02/2013 13:31	PDF File	1,587 KB
PL2303_DriverInstallerv1.8.0_ReleaseNote.txt	26/02/2013 11:10	TXT File	9 KB
PL2303_Prolific_DriverInstaller_v1.8.0.exe	26/02/2013 11:40	Application	3,136 KB
PL2303CheckChipVersion_ReadMe.txt	26/02/2013 13:38	TXT File	2 KB

2. Double-click the PL2303_Prolific_DriverInstaller_v1.8.0 executable file and follow the installation guide to install the serial port driver.

After the driver installation, if you plug in the M5X0-PWR power board and the device manager's port (COM&LPT) shows a Profile USB-to-Serial Comm Port, it indicates that the serial port is functioning correctly and can be used for normal debugging.

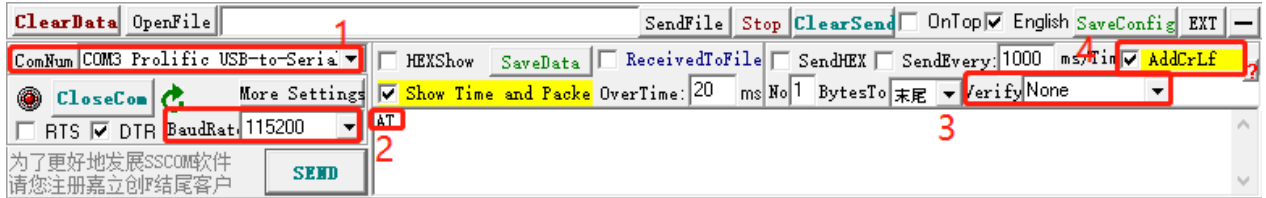


If you are using Neoway's M5X0-PWR serial port board, please contact Neoway FAE to obtain the PL2303 drivers or download them yourself.

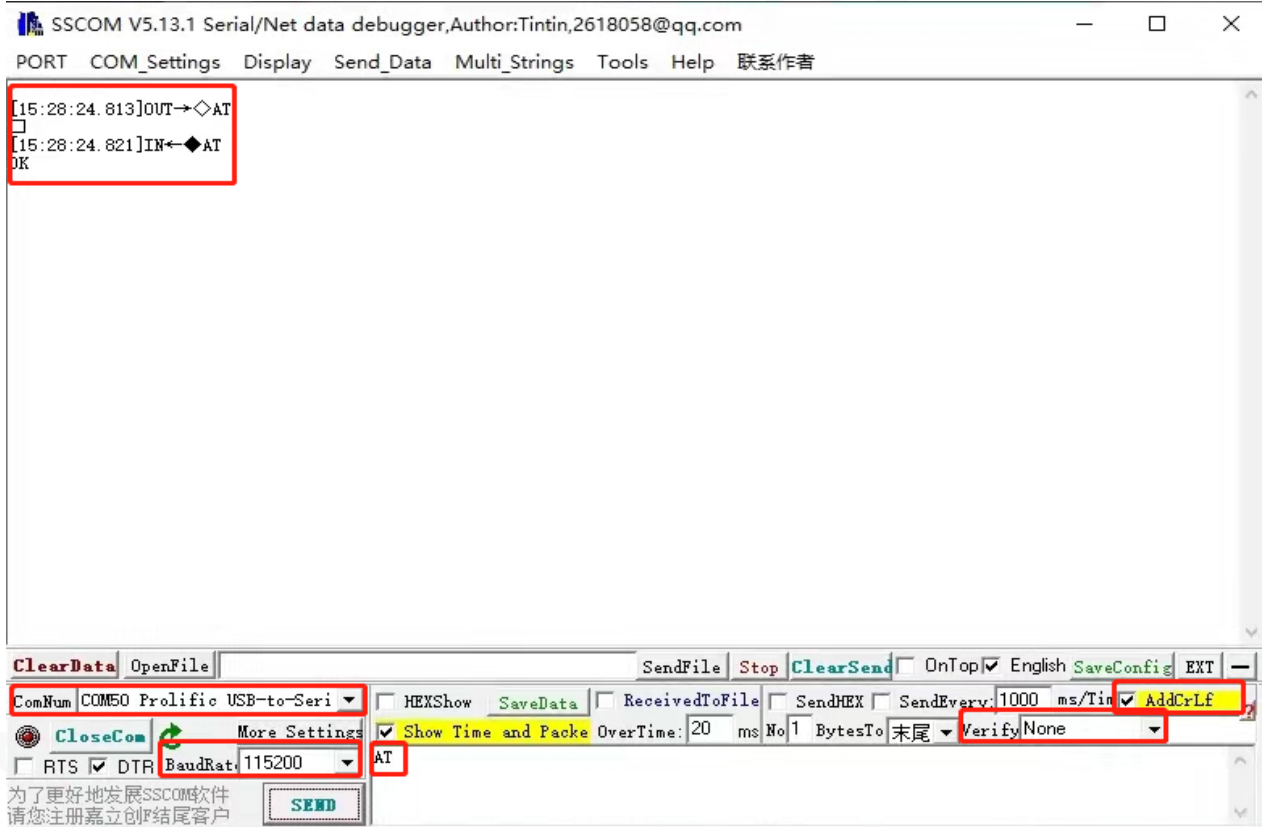
Step 5: Open the serial port tool and make the appropriate settings. Taking the serial port tool sscom as an example (users may also use other serial port tools).

The parameter settings for the serial port tool are as shown below:

1. Select the serial port number (based on the USB-to-Serial Port number displayed in the device manager interface).
2. Set the baud rate of the serial port tool to 115200 or below, for example, 115200.
3. Set parity check to None.
4. Check **AddCrLf**.



5. Send any AT command to the module using the serial port tool, such as "AT", to perform baud rate auto-adaptation. If characters are displayed in the serial port tool interface, it indicates that the module's baud rate auto-adaptation is successful.



Step 6: Use the serial tool to test or debug the module.

4.2 Debugging through USB

Step 1: Power the N706-EA EVB (See section 3.1).

Step 2: Use a Micro-USB cable to connect the computer and the EVB.

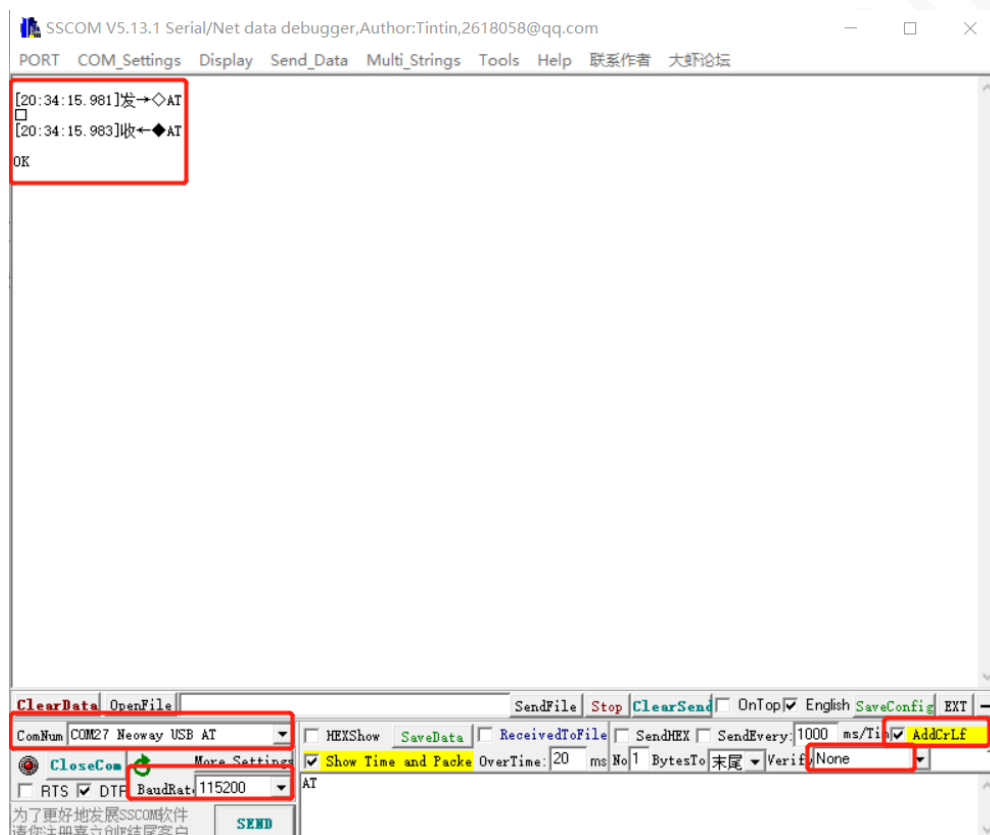
Step 3: Insert the SIM card and connect the antenna.

Step 4: Power on the module.

Step 5: Install the USB driver.

Step 6: Open the serial tool and make the corresponding settings.

The serial port number should be selected from the AT port virtually created by USB.



Step 7: Use the serial tool to test or debug the module.



Please contact Neoway FAE to obtain the latest USB driver files for the N706 module.

The USB interface of the N706-EA EVB can also be used for firmware upgrade and log capture. Please refer to the relevant application guide for specific operations.